

Solution Architecture

Date	4-7-2026
Team ID	Individual
Project Name	Loan approval prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Architecture Diagram

Presentation / Client Layer

Web Application (HTML, CSS, Bootstrap)



API Gateway

Flask Application



Core Logic Service

Data Preprocessing

Machine Learning Model (Random Forest / XGBoost)

Loan Approval Prediction



External API Integration

Email Notification (Optional)



Data / Storage Layer

MySQL Database

CSV Dataset

Trained ML Model (.pkl)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User enters loan details and views prediction result	HTML, CSS, Bootstrap
API Gateway	Handles user requests and connects frontend with ML model	Flask
Core Logic Service	Processes data and predicts loan approval	<i>Python, Scikit-learn, Pandas, NumPy</i>
Database	Stores applicant details and prediction results	MySQL, CSV